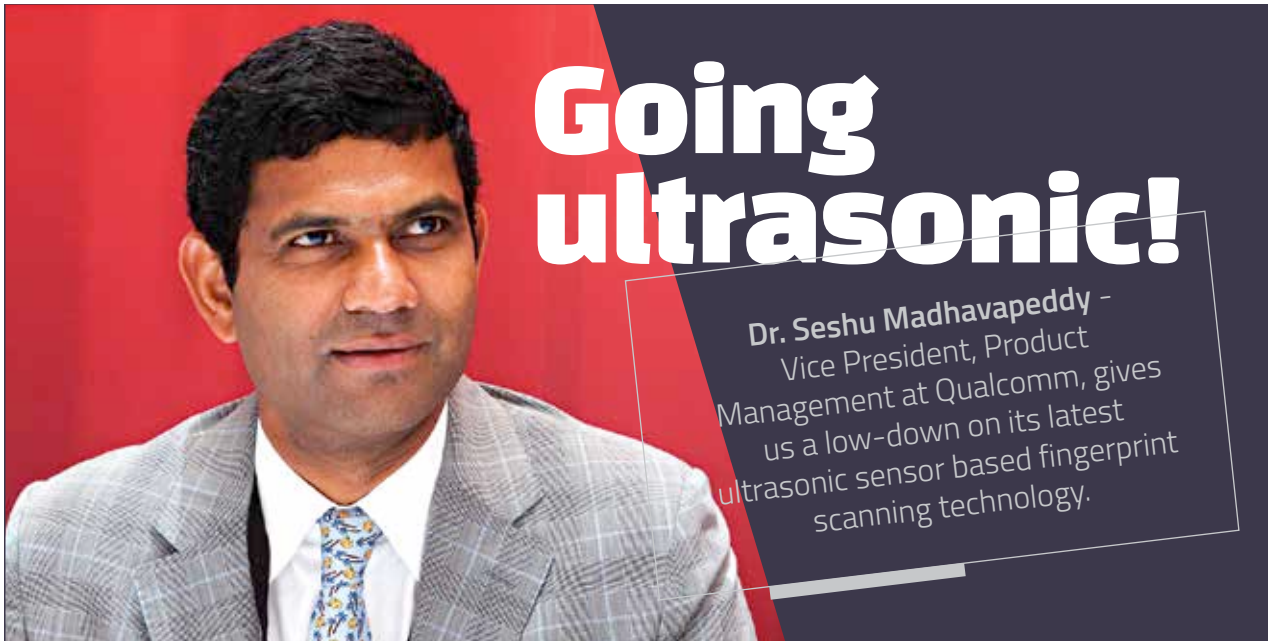




Going ultrasonic!

Dr. Seshu Madhavapeddy -
Vice President, Product
Management at Qualcomm, gives
us a low-down on its latest
ultrasonic sensor based fingerprint
scanning technology.

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How will Qualcomm's latest ultrasonic Fingerprint Solutions benefit the smartphone industry?

Dr. Seshu: QFS is sure to give users new experiences by working underwater and provide superior robustness as dirty and wet fingers too will be able to unlock a device. Additionally, it is the first commercially announced mobile solution to scan through up to 800 µm of cover glass and up to 650 µm of aluminum. Apart from its abilities to scan through various materials, it can also detect and measure heartbeat and blood flow rate. Finally, it will allow smartphone makers the freedom to experiment with designs and move towards slim bezels or even bezel-less designs.

What kind of materials and surfaces can accommodate the new finger print sensor technology?

Dr. Seshu: Since ultrasonic soundwaves can propagate through materials, it allows us to offer the technology under OLED display, under metal surfaces and even glass. The advantage being that, you don't have to cut a hole in any of the surfaces to accommodate the fingerprint sensing tech. This particularly helps the trend towards bezel less designs, where there isn't enough space to implement the finger print sensor.

Does the fingerprint solution work with all kinds of displays? And, what about detection speeds?

Dr. Seshu: The Qualcomm Ultrasonic fingerprint technology works with OLED displays. If not, the sensor can be implemented at the back or at the front, below the display, in case of standard (non-bezel less) design. As far as speeds go, the technology will offer the same speed as capacitive sensors in final commercial devices.

Do these ultrasonic sensors also offer heart rate and blood flow detection as a default feature? Will they be compatible with current health and fitness apps in the market? What about developer support?

Dr. Seshu: Heartbeat detection is performed by the same sensor as no separate sensor is needed to perform two different operations. Ultra-sonic technology enables detection of heartbeat and blood flow. It's combined with existing human tissues impedance measurement designed to offer powerful liveness detection, and improved mobile authentication. Yes, the technology will be compatible with all popular fitness and health apps with no modification needed. The developers do not have to worry about what type of a fingerprint sensor is being used on a device. They can use standard API offered by the Android operating system to be able to access the fingerprint sensor function.

How many OEMs are working with this solution and by when can we expect phones with this tech? Also, does an OEM compulsorily need to use Qualcomm Snapdragon platform to use the QFS solution?

Dr. Seshu: We have just introduced the solution in July, and we can expect phones with this tech to start coming in the beginning of next year. Commercial devices with fingerprint sensors for display are expected to arrive in summer of 2018, while commercial devices with fingerprint sensors for glass and metal are expected to arrive in early 2018. Qualcomm is offering this tech as a standalone solution, OEMs don't have to use Snapdragon platform to implement the solution.

What about power efficiency and security of this technology? How does it compare to the current finger print sensor tech?

Dr. Seshu: In terms of power consumption, the device is as power efficient as the current finger print sensor technologies. As far as security goes, we can deliver higher security levels via our ultrasonic sensors in comparison to capacitive sensors. For instance, our ultrasonic sensors can detect if the finger is a live finger or a fake finger with the same imprint, which is not the case when it comes to capacitive technology based sensors. 